

## FINAL EXAMINATION

|                                                          |                                         |
|----------------------------------------------------------|-----------------------------------------|
| Course: <b>SC203 – Scientific Research Methodologies</b> |                                         |
| Time: <b>110</b> minutes                                 | Term: – Academic year: <b>2020-2021</b> |
| Lecturer(s):                                             |                                         |
| Student name:                                            | Student ID:                             |

*(Notes: Books, laptops, document are all allowed)*

**Quiz:** (4 marks) (*Students select only one correct choice in each question*)

**Q.1:** Which is **NOT** a cause of random errors?

- A.** Random processes within system
- B.** Limitations of measuring tool
- C.** Observer reading output of tool
- D.** Repeated measurements

**Q.2:** Which is **NOT** an experimentation strategy?

- A.** Fractional Factorial
- B.** Educated guessing
- C.** Proportionate stratified
- D.** One-factor-at-a-time

**Q.3:** Which is **NOT** a purpose of research proposal?

- A.** To explain why the research is needed.
- B.** To explain the nature of the research
- C.** To explain the context of the research
- D.** To clearly state the exact outcomes of the research.

**Q.4:** Which is **NOT** an aim of doing experiments?

- A.** To take a detached view of the phenomena, and be ‘invisible’, either in fact or in effect.
- B.** To find out what happens if you make a change
- C.** To isolate a particular event so that it can be investigated without disturbance from its surroundings.
- D.** To gain data about causes and effects of a particular event

**Q.5:** Which is **INCORRECT** in stages of research project?

- A.** Literature review.
- B.** Analysis of evidence.
- C.** Hypothesis.

**D.** Technical report does not determine the scope of hypothesis.

**Q.6:** In a scientific paper, you need to:

**A.** Justify the theory, by methods such as proof or experiment.

**B.** State the idea that is being investigated, often as a theory or hypothesis.

**C.** Explain what is new about the idea, what is being evaluated, or what contribution the paper is making.

**D.** All of the above

**Q.7:** Which of the following statements is **FALSE**?

**A.** The main reason that we experiment, and measure is to provide evidence about the behavior of a system in general—not just on some specific data set.

**B.** When considering what experiments to try, you should identify the data or input for which the hypothesis is most likely to hold.

**C.** When checking experimental design or outcomes, consider whether there are other possible interpretations of the results; and, if so, design further tests to eliminate these possibilities.

**D.** Experiments in computing take diverse forms, from tests of algorithm performance to human factors analysis.

**Q.8:** Why should researchers keep detailed notebooks?

**A.** They allow easy reconstruction of old research.

**B.** They simplify the process of write-up.

**C.** They keep researchers honest.

**D.** All of the above.

**Q.9:** Asking questions is one of the basic methods used to collect primary data. The advantage(s) of using this method is that:

**A.** It is easy and convenient for respondents. **B.** Probe questions can be asked.

**C.** Both A and B. **D.** Complex question structure are not possible.

**Q.10:** What is the level of measurement for temperature?

**A.** Nominal

**B.** Ordinal

**C.** Interval

**D.** Ratio

**Q.11:** Which is the necessary characteristic for research from the beginning?

A. Intelligence.

B. Curiosity.

C. Kindness.

D. Honesty.

**Q.12:** The key principle of describing experiments is that:

A. You should distinguish what has been done in the field from what needs to be done.

B. Reported results should be a fair reflection of the experiment's outcomes.

C. The experiment should be verifiable and reproducible.

D. You should place the topic or problem in the broader scholarly literature.

**Q.13:** What **DOES NOT** need to be cited?

A. Claims

B. Common knowledge

C. Discussion of previous work

D. Statements of fact

**Q.14:** Which of the following statements is correct about hypotheses of research projects?

A. Where two hypotheses fit the observations equally well and one is clearly simpler than the other, the simpler should be chosen.

B. A hypothesis must be capable of falsification.

C. A hypothesis should lead to predictive results.

D. All of the above.

**Q.15:** What are the basic principles of statistical design of experiments?

A. Randomization, Replication, Blocking

B. Replication, Blocking, Interaction

C. Randomization, Optimization

D. Replication, Confirmation

**Q.16:** When each element of a population has an equal chance of being selected, this is called:

A. Probability sampling

B. Non-probability sampling

C. Snowball sampling

D. Accidental sampling

**Q.17:** What is the purpose of literature review?

A. To know the current state of knowledge in the subject

B. To assess the current state of knowledge for relevance, quality, controversy and gaps

C. To resolve a controversy or fill a gap

D. Both A and B

**Q.18:** Which of the following statements is **FALSE**?

A. Order, external reality, reliability, parsimony, and generality are the five assumptions that underlie scientific method.

B. The essential nature of a hypothesis is that it must be impossible to falsify.

C. Science is seen to proceed by trial and error: when one theory is rejected, another is proposed and tested, and thus the fittest theory survives.

D. The hypothetico-deductive method, which combines experience with deductive and inductive reasoning, is the foundation of modern scientific research and is commonly referred to as scientific method.

**Q.19:** What is the main purpose of academic research?

A. To advance the frontiers of human knowledge

B. To explore the data against new dimensions

C. To summarize the characteristics of a data set

D. To analyse negative cases

**Q.20:** You should choose a form of evidence used to support your hypothesis so that:

A. The evidence is as persuasive as possible.

B. It keeps your own effort to a minimum.

C. It makes your paper complicated and difficult to understand.

D. All of the above.

**ESSAYS:** (6 marks)

Given a draft of paper in the following pages, please help the author to

1. Select a list of 5 keywords from the paper (1 mark)
2. Write the title of the paper (1.5 marks)
3. Write the abstract of the paper: 8 - 10 sentences (2 marks)
4. Write the conclusion of the paper: 6 – 8 sentences (1.5 mark)

# Title

Name\*

\* University of Science, HCM City, Vietnam

*Abstract—*  
*Index Terms—*

## I. INTRODUCTION

Many applications such as predicting gold price or predicting stock price can be formed as time series forecasting problems [1]. These applications' data are usually sequences of numbers or vectors corresponding with time. The fluctuated range of data can be vary depending on the kind and properties of data. In a time series forecasting problem, the primary mission is to predict the next value in the series based on the values in the past.

Time series forecasting [2] [3] is very useful in many real-world applications including economic, management, environment, healthcare, stock prediction [4] [5], etc. This problem means constructing a model from available data to predict the next values in the near future. If the data can be analyzed with rules, the predicted values will be approximated with the truth or the value in the real world. These results should be applied to the world to adjust human behaviours and then gain a better state. The results of the forecasting model provide a quantitative method to measure the impact of past and present on the future.

Forecasting is not an easy problem [6]. There are at least two agents that make the predicting process is so complicated. The first one is whether the predicted value can be predicted. It cannot predict that whether an Adidas store will sell off tomorrow. That event mostly depends on the store owner's decision or the store policy. Many forecasting methods focus on finding the rules or the trends in the data to make a guest, so these methods could not work well with the random data or emotion-based data. The second aspect is the ability to collect enough data to build the predicting model. Data should be collected enough in terms of amount and related context. While the amount can be satisfied, the data context is much more challenging to collect and organize. How many things affect the gold price is a tricky question. On the other hand, the way we structure a database with its own contexts is complicated. These two reasons, especially the second one, mainly cause the difficulty of time series forecasting.

We propose a simple method to predict the following values for time series data in this work. Our approach focuses on the comparative periodic data, which is a common data type in real-world applications. With this type, data is followed by some rules or trends. Remarkably, these data are repeated during the time. We exploit these rules with an integral transform [7]. The transform, in nature, is a mathematical way to separate an occasional series into many periodic simple

series called basic. The set of basic functions can present all the rules and trends in the data, and then apply these factors to compute the values in the future. With a cheap computing cost, this method can forecast the next value for the nearly repeated data very efficiently.

The remaining of this paper includes four sections. Section two will present the general approach for the problem of time series forecasting. The next section shows our proposed method. This section presents our transform and its application to calculate the next value in the series. Section four is the application of our data transform to sports data. Particularly, we apply the transform method to sports data to exploit its trend. The final section will summarize our work and then make a conclusion.

## II. A GENERAL FRAMEWORK FOR TIME SERIES FORECASTING

To deal with a forecasting problem, there are three main stages that we should complete:

- Defining the problem and collecting data: Identifying what the model will predict, data volume, data properties, and how to collect data.
- Building model: Identifying the input and output of the model, what the main processing model should be.
- Predicting the next values: Specifying the way to apply the constructed model to predict the values in the near future.

In the three stages above, the second stage, building model, can be the most important step. The accuracy and performance of the final stage mostly depend on the quality of the built model in this stage. Constructing a right and efficient model will learn and present the data rules very well. This is fundamental for a good prediction.

## III. DATA TRANSFORMATION AND ITS APPLICATION TO DATA FORECASTING

In many applications, an original format of data is difficult to analyse or process. We need to transform the data into another form to become more informative. There are many techniques for data processing in general, and integral transform is one of the most popular choices.

Integral transform [7] [8] is a mathematical tool to transform data from this form to another form via an integral operator. Let  $f(t)$  and  $K(t, u)$  denote the original data, and the kernel, the integral with the kernel  $K(\cdot)$  can be defined as:

- Forward transform:

$$Tf(u) = \int_{t_1}^{t_2} f(t)K(t, u)dt \quad (1)$$

TABLE I  
THE CONCEDED GOAL/MATCH RATE OF THE TEAMS COACHED BY PEP  
GUARDIOLA

| Team          | Year      | Match | Conceded goal | Conceded goal/n |
|---------------|-----------|-------|---------------|-----------------|
| Barcelona     | 2008-2009 | 38    | 35            | 0.92            |
| Barcelona     | 2009-2010 | 38    | 24            | 0.63            |
| Barcelona     | 2010-2011 | 38    | 21            | 0.55            |
| Barcelona     | 2011-2012 | 38    | 29            | 0.76            |
| Bayern Munich | 2013-2014 | 34    | 23            | 0.68            |
| Bayern Munich | 2014-2015 | 34    | 18            | 0.53            |
| Bayern Munich | 2015-2016 | 34    | 17            | 0.50            |
| Man. City     | 2016-2017 | 38    | 39            | 1.03            |
| Man. City     | 2017-2018 | 38    | 27            | 0.71            |
| Man. City     | 2018-2019 | 38    | 23            | 0.61            |
| Man. City     | 2019-2020 | 38    | 35            | 0.92            |
| Man. City     | 2020-2021 | 38    | ?             | ?               |

- Inverse transform:

$$f(t) = \int_{u_1}^{u_2} T f(u) K^{-1}(t, u) du \quad (2)$$

Forward transform is used to explore the hidden properties of the data in the other space. The inverse transform is applied to reconstruct the data after being processed in the other space. Generally, we use all these transforms in reality for many purposes.

#### IV. TRANSFORM AND FORECASTING IN SPORT DATA

We use the integral transform for analyzing the sport data. Particularly, we consider some football teams including Barcelona, Bayern Munich, and Manchester City, which Pep Guardiola coaches. Using the proposed method, we try to predict the result of Pep's team via predicting the average conceded goal per match.

##### A. Analyzing data

Pep Guardiola is a genius in football, especially with the role of a coach. His teams always have a clear brand identity due to their playing styles. Pep inherits the style from the traditional Netherlands style and develops it to reach a new level. Pep and his teams won and lost due to his strategy for every match.

Pep usually spends from three to five years with each team. In the final years in the teams, his achievements are so bad, and then he would be fired. One or two years before, his results were much better and win many champions. On the other hand, the results in the first years are not good compared to the second year.

One of the most aspects, which affects directly to the performance of a team, is the ratio of conceded goals per match. When the teams perform well and win some champions, this ratio is low. If this ratio is high, the teams play too badly and lose too much. The achievements of Pep is present in the table I with the rightmost column. The number for the years from 2008 to 2019 are given due to the statistics while the number for this year (2020-2021) is still a question. The forecasting problem means predicting the value for this year. For more visibly, these values are illustrated in the figure 1.

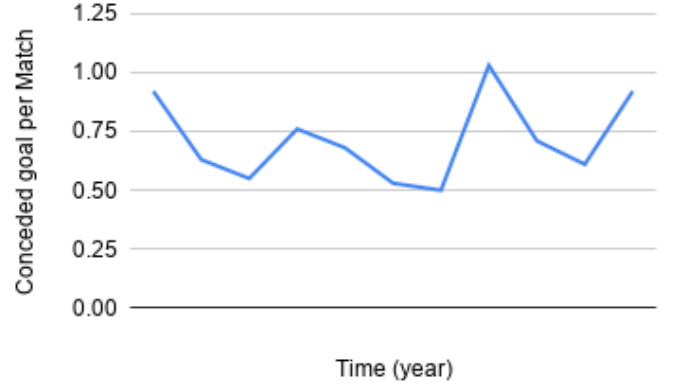


Fig. 1. The average conceded goal per match of Pep's team.

It is easy to observe in the figure 1, the distribution of conceded goal is near a periodic waveform. Pep's achievements can be separated into three-stage corresponding with three teams and three periods in the figure 1. The local bottom values are with the third, seventh, and tenth years. They locate at the two to three years in each team and are the most successful year of these teams. This fact can be interpreted by analyzing Pep's strategy as follow:

- Mostly depending on unique techniques, not strong or power.
- Needing the time for the player to adapt with the strategy.
- First year: Not all player plays well because they need more time.
- The second to the fourth year: The team is formed, they play well and perform full strengths of the strategy.
- The final year: The opponents are familiar with the strategy and counter-work to defeat Pep's team.

##### B. Framework for forecasting

As presented in the previous subsection, Pep's teams are usually followed by a fixed strategy and form a fixed achievement curve with three to five years in terms of period. This aspect is very fitted to the strength of Fourier transform. Thus, in this research, we apply this mathematical tool to exploit the data. We apply the below framework for this process:

- Modeling phase: In this phase, we transform the data into the frequency domain via Discrete Fourier Transform [9] [10] with data  $f(t)$  and eleven values for  $\omega_k$ :

$$F(\omega_k) = \frac{1}{2\pi} \sum_{t=1}^{11} f(t) e^{-i\omega_k t} \quad (3)$$

- Forecasting phase: Applying Inverse Discrete Fourier Transform to compute the next value in  $f(t)$ :

$$f(t) = \frac{2\pi}{11} \sum_{k=1}^{11} F(\omega_k) e^{i\omega_k t} \quad (4)$$

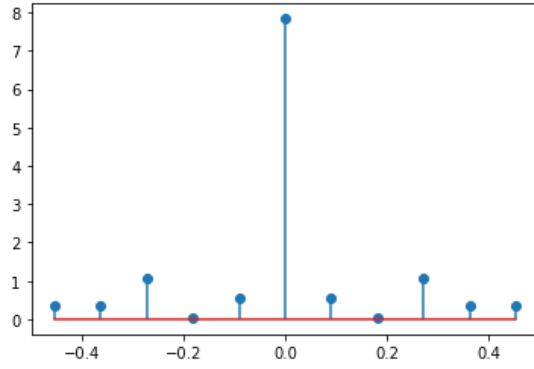


Fig. 2. Frequency representation for Pep's achievement

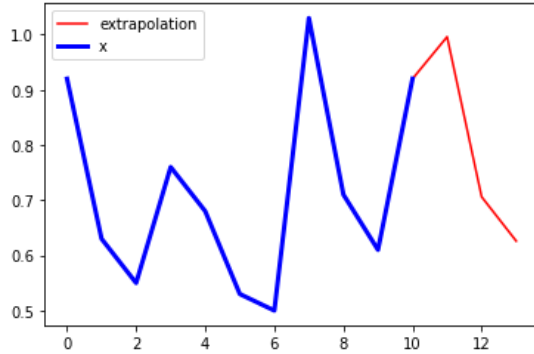


Fig. 3. Predicted value for Pep's team in the next three years

### C. Experimental result

In the modelling phase, the frequency coefficients are presented in the figure 2. These elements correspond with the eleven basic waveforms of the data. As can be seen, these values distribute very differently from the original data during the time.

Figure 3 shows the Forecasting phase. After computing the next values in the data series, the results are:

$$f(12) = 0.996 \quad (5)$$

$$f(13) = 0.706 \quad (6)$$

$$f(14) = 0.626 \quad (7)$$

This means that the model predicts Pep's team will not win a lot in 2021 and can lose so much. Let us check the results in reality and then compare with the result showed by the model.

At the end of 2020-2021 season, Pep's team plays 38 matches, and concedes 32 goals. So we have:

$$\hat{f}(12) = 32/38 = 0.842 \quad (8)$$

The season 2021-2022 are continuing (at Dec, 2021) so we can use the current result as a representative. The team has

played 20 matches and received 12 conceded goals. So we can estimate:

$$\hat{f}(13) = 12/20 = 0.600 \quad (9)$$

As can be observed, the numbers predicted by our method and their real values are different, but there are two things meaningful in the results. The first is that the differences not too significant. The results can be accepted in a prediction task if the distance from an estimated value to its real value is small enough. The second, and more important, is the trend of sequence value. From  $f(12)$  to  $f(13)$ , in the values estimated by our model, the decreasing rate is:

$$d_r = 1 - \frac{0.996}{0.706} = 0.411 \quad (10)$$

On the other hand, this rate in reality is computed as:

$$\hat{d}_r = 1 - \frac{0.842}{0.600} = 0.403 \quad (11)$$

The trends are decreasing, and the decreasing rates are too similar. This is evidence to prove that our method can exploit many hidden behaviours in the data and predict the values and the trend in the time series data.

## V. CONCLUSION

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